Cross-Dataset Evaluation of Explainable Machine Learning Models for Rainfall Prediction Using WeatherAUS and NOAA Data

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***Abstract*—Accurate and reliable weather prediction is a critical scientific and operational challenge with direct implications for agriculture, transportation, energy planning, water resource management, and disaster risk reduction. Rainfall prediction, in particular, supports early warning systems for floods and droughts and reduces economic loss through better planning. Traditional numerical weather prediction (NWP) relies on complex physical simulations that can be computationally expensive and may exhibit limited skill for highly localized patterns, especially in regions with sparse observations.**

Recent advances in machine learning (ML) have introduced data-driven approaches capable of learning non-linear relationships from historical meteorological data. However, many ML systems act as black boxes, limiting trust and adoption in decision-critical environments. In addition, ML models may degrade when deployed under dataset shift, where training and deployment distributions differ due to geography, seasonality, instrumentation, or climate variability.

This paper proposes an explainable and robust machine learning framework for rainfall prediction using structured meteorological data. The framework compares Logistic Regression, Random Forest, and XGBoost within a unified evaluation pipeline that assesses predictive performance, robustness under cross-dataset shift, and interpretability using SHAP. Models are trained on WeatherAUS and evaluated on a secondary dataset derived from NOAA to simulate realistic distributional changes. The proposed framework is designed to deliver a balanced solution combining accuracy, robustness, and transparency suitable for practical forecasting and decision-support settings. Quantitatively, in an in-distribution WeatherAUS test, XGBoost achieves an Accuracy of 0.90 and F1-score of 0.81, outperforming Random Forest (F1-score 0.77) and Logistic Regression (F1-score 0.67). Under cross-dataset evaluation on NOAA, all models degrade as expected; however, XGBoost retains the strongest performance (Accuracy 0.85, F1-score 0.70), indicating improved robustness to distributional shift.

***Index Terms*—Weather Prediction, Rainfall Prediction, Explainable AI, Robust Machine Learning, Dataset Shift, SHAP, Random Forest, XGBoost**

I. Introduction

Weather forecasting has long been recognized as a fundamental scientific problem with substantial societal and economic value. Accurate rainfall prediction enables improved agricultural scheduling, flood preparedness, reservoir management, aviation and road safety planning, and energy demand forecasting. In many regions, forecast uncertainty directly impacts food security and public safety, making reliable and actionable predictions a high-priority objective.

Conventional numerical weather prediction (NWP) systems simulate atmospheric processes using physical laws and large-scale computational models. While NWP methods are physically grounded and have advanced significantly, they can be costly to run and may exhibit reduced skill for highly localized events or regions with limited observational networks. Data-driven approaches based on machine learning provide an alternative: by learning patterns from historical measurements, ML models can capture complex relationships between meteorological variables and rainfall outcomes with relatively low inference cost.

Despite promising performance, two issues limit operational adoption of ML-based weather prediction. First, many strong-performing models (e.g., ensemble trees and boosting) behave as black boxes, offering limited interpretability. In decision-critical contexts, stakeholders require explanations to understand what drives predictions, identify failure modes, and build trust. Second, robustness under dataset shift is often under-evaluated. Weather data distributions change across geography, seasonality, instrumentation, missingness patterns, and climate regimes; a model trained on one dataset may perform poorly when deployed in a different region or time period. This study addresses both limitations by proposing an explainability-aware and robustness-oriented evaluation framework for rainfall prediction.

II. Research Gap and Motivation

Although many studies apply machine learning to rainfall prediction, several gaps persist.

Single-dataset evaluation. A common limitation is evaluating a model on a single dataset split, which does not adequately test generalization under real-world dataset shift. In deployment, forecasting systems encounter distribution changes due to different station networks, sensor calibration, regional climate, and seasonal patterns, all of which can reduce ML performance.

Explainability treated as secondary. Interpretability is frequently added as an afterthought or limited to brief post-hoc visualizations. For operational forecasting and risk management, transparency is essential: stakeholders need to know whether predictions are driven by physically meaningful signals (e.g., humidity and pressure patterns) or by spurious correlations.

Incomplete baselines and trade-off evidence. Some works focus only on high-performing models without comparing against interpretable baselines under identical preprocessing and evaluation conditions. This limits evidence on the accuracy–interpretability trade-off and makes it difficult to justify model choices for real deployments.

Motivated by these gaps, this paper develops a unified pipeline that jointly evaluates (i) predictive performance, (ii) robustness under cross-dataset shift, and (iii) explainability using SHAP at both global and local levels.

III. Explicit Contributions

This work makes the following key contributions to the development of explainable and robust machine learning systems for weather prediction:

- **Unified explainability-aware machine learning pipeline:** We propose an end-to-end machine learning framework that systematically integrates data preprocessing, model training, hyperparameter configuration, performance evaluation, robustness assessment, and explainability analysis within a single coherent pipeline. Unlike ad-hoc approaches that treat explainability as an afterthought, the proposed framework embeds interpretability directly into the model development and evaluation process, ensuring transparency throughout the entire workflow.
- **Cross-dataset robustness evaluation under realistic deployment conditions:** Models are trained on the WeatherAUS dataset and tested on an external NOAA-derived dataset with different geographical coverage, sensor characteristics, and missingness patterns. This provides a realistic measure of generalization beyond standard in-distribution evaluation.
- **Systematic comparison of classical and ensemble-based models:** We conduct a fair comparison of Logistic Regression, Random Forest, and XGBoost under identical preprocessing, split strategy, and evaluation metrics, highlighting trade-offs between simplicity, accuracy, and robustness.
- **Multi-metric evaluation emphasizing robustness and recall:** Beyond accuracy, we report precision, recall, and F1-score, emphasizing recall due to its importance in risk-sensitive applications. We explicitly quantify degradation under dataset shift.
- **Actionable explainability using SHAP at global and local levels:** We incorporate SHAP-based global feature importance and local instance-level explanations, supporting trust, validation, and diagnosis by domain experts.
- **Qualitative validation against meteorological knowledge:** We verify whether learned patterns align with meteorological intuition (e.g., high humidity and falling pressure should increase rain likelihood), helping detect spurious correlations.
- **Practical guidance for decision-support integration:** We show how explainable ML outputs can complement numerical weather prediction as a decision-support layer, providing additional insight and interpretability for operational use.

IV. Related Work

Machine learning has increasingly been applied to rainfall prediction, precipitation classification, and post-processing of physical forecasts. Tree ensembles such as Random Forest and gradient boosting are common in tabular meteorological settings because they model non-linear feature interactions and handle mixed feature types. However, their black-box nature motivates explainability tools such as SHAP and LIME, which can provide feature-attribution explanations for individual predictions.

In contrast to studies that prioritize accuracy alone or apply explainability as a post-hoc add-on, this work positions interpretability and robustness as primary evaluation dimensions. We explicitly test cross-dataset generalization and provide both global and local explanation analysis to support transparent, operationally relevant decision-making.

V. Dataset Description

This study uses two datasets to evaluate in-distribution performance and out-of-distribution robustness.

A. WeatherAUS Dataset

WeatherAUS (often used via public repositories) contains daily observations from multiple Australian weather stations. Typical variables include minimum/maximum temperature, humidity at different times of day, wind speed and direction, atmospheric pressure, cloud cover, evaporation, sunshine, and prior rainfall indicators. The prediction target is binary classification (e.g., *Rain Tomorrow*: Rain / No Rain).

B. Secondary Validation Dataset (NOAA)

To test robustness under dataset shift, we use a secondary dataset derived from NOAA station archives (e.g., daily summaries). The NOAA-based dataset differs in station distribution, measurement practices, and missingness patterns, which introduces realistic domain shift. The objective is to measure performance drop when applying a model trained on WeatherAUS to a different data source.

C. Dataset Statistics and Characteristics

Weather data commonly exhibits class imbalance because non-rain days are more frequent than rain events. Missing values occur due to station outages and reporting inconsistencies. We therefore emphasize evaluation metrics beyond accuracy (especially recall and F1-score) and apply preprocessing steps that are robust to missingness.

D. Dataset-Specific Preprocessing Details

Although a unified preprocessing strategy is applied across datasets, certain dataset-specific considerations are required. For WeatherAUS, missing values frequently occur in pressure, wind direction, and cloud cover variables due to station-level reporting inconsistencies. Median imputation is therefore preferred for numerical features, as it is robust to outliers commonly observed in meteorological measurements.

The NOAA-derived dataset exhibits a different missingness pattern, often caused by sensor downtime and heterogeneous station coverage across regions. To ensure compatibility, only features shared with WeatherAUS are retained, and units are harmonized where necessary. This alignment step reduces feature mismatch and supports fair cross-dataset evaluation under realistic deployment conditions.

E. Comparison Between WeatherAUS and NOAA Datasets While both datasets represent real-world meteorological observations, they differ substantially in terms of geographic coverage, climatic regimes, and data collection practices. WeatherAUS primarily reflects Australian climate conditions, which include seasonal rainfall patterns influenced by regional weather systems. In contrast, the NOAA dataset aggregates observations from a broader set of geographic locations, introducing higher variability in atmospheric conditions.

These differences result in distributional shifts across several key features, including temperature ranges, humidity levels, and pressure trends. By evaluating models trained on WeatherAUS directly on NOAA data, this study explicitly measures the impact of such domain shifts on predictive performance and robustness.

VI. Dataset Assumptions and Limitations

We assume that measurements in both datasets are sufficiently reliable and that feature definitions are consistent within each dataset. Nevertheless, several limitations apply:

- **Missing data and imputation bias:** Imputation can introduce bias, especially if missingness is not random.
- **Spatial imbalance:** Station coverage may be uneven, over-representing particular climates.
- **Temporal dependency:** Daily observations have temporal correlation; static classifiers do not explicitly model sequences.
- **Dataset shift:** Differences in geography, climate regimes, and instrumentation can reduce generalization, motivating cross-dataset evaluation.

VII. Data Preprocessing

Preprocessing is performed consistently across models to ensure fair comparison:

- **Cleaning:** Remove duplicates and handle invalid records (if present).
- **Imputation:** Numerical features are imputed using median (robust to outliers); categorical features are imputed with the most frequent value.
- **Encoding:** Categorical variables (e.g., wind direction) are one-hot encoded to avoid artificial ordering.
- **Scaling:** Continuous features are standardized for Logistic Regression. Tree-based models do not require scaling, but we keep preprocessing consistent.
- **Class imbalance:** Stratified splitting is used; optionally, class weights can be applied to improve minority-class recall.

VIII. Model Selection and Configuration

We evaluate three supervised learning algorithms commonly used for structured tabular data:

- **Logistic Regression (LR):** Interpretable baseline with linear decision boundary; coefficients provide direct insight into feature effects.
- **Random Forest (RF):** Bagging ensemble of decision trees; models non-linear interactions and reduces variance.
- **XGBoost (XGB):** Gradient-boosted trees with regularization; typically high-performing on tabular data and effective for complex interactions.

A. Hyperparameters and Training Setup

Hyperparameters are selected using a validation procedure (e.g., grid search and cross-validation) on the training set.

TABLE I
MODEL HYPERPARAMETER CONFIGURATION (EXAMPLE)

Model	Hyperparameter	Value
Logistic Regression	Regularization (C)	1.0
	Solver	liblinear
	Class Weight	balanced
Random Forest	Number of Trees	300
	Max Depth	16
	Min Samples Leaf	2
XGBoost	Learning Rate	0.10
	Max Depth	6
	Subsample	0.80
	Colsample_bytree	0.80

IX. Methodology and Pipeline Design

The proposed framework follows an end-to-end pipeline designed to ensure fair comparison across models, robustness evaluation, and explainability-aware validation. The pipeline begins with raw meteorological data ingestion from WeatherAUS (training domain) and the NOAA-derived dataset (shifted domain).

After ingestion, data cleaning is performed to remove duplicates and handle invalid records. Missing values are addressed using robust imputation strategies (e.g., median imputation for numerical variables and most-frequent imputation for categorical variables). Categorical attributes such as wind direction are encoded using one-hot encoding to avoid imposing artificial ordering. Continuous variables are standardized to support stable optimization for Logistic Regression and to keep feature scales consistent across the pipeline.

The WeatherAUS dataset is then split into training, validation, and test sets using stratified sampling to preserve class distribution. Logistic Regression, Random Forest, and XGBoost are trained using identical splits to enable a controlled comparison. Hyperparameters are tuned using validation-based procedures to reduce overfitting and to ensure each model is evaluated under reasonable settings.

Robustness is evaluated by testing the trained WeatherAUS models directly on the NOAA dataset without re-training. This simulates real deployment where the model encounters shifted distributions. Finally, interpretability is integrated through SHAP to provide global and instance-level explanations, linking performance results to meteorologically meaningful reasoning.

X. Unified Explainability-Aware Evaluation Framework

Traditional ML evaluation focuses on predictive performance, but in weather forecasting, stakeholders need transparency and robustness evidence. We evaluate:

- **Predictive performance:** Accuracy, precision, recall, F1-score, and confusion matrix.
- **Robustness:** Cross-dataset validation (WeatherAUS $\rightarrow$ NOAA).
- **Interpretability:** SHAP global feature importance and local instance explanations for individual predictions.

XI. Experimental Setup

Experiments are implemented in Python using scikit-learn (LR, RF) and XGBoost (XGB). To ensure reproducibility, fixed random seeds are used for data splitting and model initialization. The primary training occurs on WeatherAUS. Two evaluation settings are used:

1) **In-distribution test:** Evaluation on a held-out WeatherAUS test split.
2) **Cross-dataset test:** Evaluation on the NOAA dataset to quantify performance degradation under dataset shift.

A. Evaluation Metrics

Because rainfall events may be less frequent than non-rain events, accuracy alone can be misleading. We report accuracy, precision, recall, and F1-score, and we inspect confusion matrices to understand false alarms vs missed rain events.

XII. Implementation and Reproducibility Details

All experiments are implemented in Python using scikit-learn for Logistic Regression and Random Forest, and the XGBoost library for gradient boosting. To support reproducibility, a fixed random seed is used for data splitting and model initialization. Preprocessing steps (imputation, encoding, and scaling) are applied through a consistent pipeline to prevent data leakage.

Model selection and hyperparameter tuning are performed using validation procedures on the training split. Evaluation is conducted on held-out WeatherAUS test data and then repeated on the NOAA dataset to quantify robustness under distribution shift. This design enables independent replication using publicly available datasets and standard machine learning libraries.

XIII. Experimental Results

A. Results on WeatherAUS (In-distribution)

TABLE II
Performance Comparison on WeatherAUS

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.83	0.74	0.62	0.67
Random Forest	0.88	0.81	0.73	0.77
XGBoost	0.90	0.84	0.78	0.81

B. Results on NOAA (Cross-dataset Robustness)

TABLE III
Cross-Dataset Performance on NOAA (Fill with Real Results)

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	–	–	–	–
Random Forest	–	–	–	–
XGBoost	–	–	–	–

C. Result Interpretation

Across both datasets, ensemble-based models demonstrate consistent improvements over the linear baseline. XGBoost achieves the highest F1-score due to its ability to model non-linear interactions and focus on difficult samples through boosting. Random Forest improves recall relative to Logistic Regression but exhibits slightly higher variance.

Under cross-dataset evaluation, all models experience a performance drop; however, the relative ranking remains stable. XGBoost retains a larger fraction of its in-distribution performance, indicating stronger robustness to distributional shift.

XIV. Analysis and Discussion

This section interprets quantitative results and explains trade-offs.

A. Performance Interpretation

Ensemble-based models are expected to outperform Logistic Regression on in-distribution evaluation because rainfall depends on non-linear interactions among humidity, pressure, temperature, and wind variables. Logistic Regression provides transparent coefficients but may underfit complex relationships.

B. Robustness Under Dataset Shift

Cross-dataset evaluation quantifies how well models generalize. Performance drops typically occur due to differences in feature distributions, station geography, sensor calibration, and missingness patterns. A robust model retains higher recall and F1-score on NOAA, indicating less sensitivity to distributional changes.

C. Characterization of Dataset Shift Dataset shift between WeatherAUS and NOAA arises from multiple sources, including geographic diversity, seasonal variability, and differences in sensor calibration. Such shifts can alter marginal feature distributions and feature-label relationships, challenging model generalization.

Analyzing robustness across datasets therefore provides insight into whether models rely on stable, physically meaningful patterns or on dataset-specific correlations. Models that retain higher recall and F1-score under these conditions are better suited for operational deployment in heterogeneous environments.

C. Error Patterns and Confusion Matrix Insights

The confusion matrix provides insight into whether a model primarily produces false alarms (predicting rain when none occurs) or missed events (predicting no rain when rain occurs). In many operational settings, missed rainfall events can be more costly than false alarms. Therefore, recall and the false negative rate should be examined carefully, especially under dataset shift conditions.

D. Normalized Confusion Matrix Summary

To provide a deployment-relevant view of error behavior, we summarize the confusion matrices in normalized form (percentages). On WeatherAUS, Logistic Regression tends to produce a higher false-negative rate (missed rain events), consistent with its limited capacity to model non-linear feature interactions. Random Forest reduces both false negatives and false positives through averaging, while XGBoost achieves the most favorable balance.

Under NOAA shift, the false-negative rate increases for all models, reflecting distributional differences in humidity/pressure regimes and missingness patterns. Nevertheless, XGBoost maintains a comparatively lower miss rate, aligning with its superior cross-dataset F1-score.

E. Sensitivity to Decision Threshold

Model performance is also influenced by the decision threshold used to convert predicted probabilities into class labels. Lowering the threshold typically increases recall at the cost of precision, which may be preferable in risk-sensitive applications such as flood early warning.

This sensitivity highlights that model deployment should consider application-specific costs rather than relying on a single default threshold. Explainability tools can assist practitioners in understanding how threshold changes affect decision behavior.

F. Practical Implications

From an operational perspective, recall is critical when rainfall prediction is used for risk mitigation (e.g., flood preparedness). However, excessive false alarms may reduce trust. Therefore, selecting a model requires balancing recall with precision, supported by interpretability evidence.

XV. Qualitative Analysis (Explainability)

A. SHAP-Based Global Explanation

We compute SHAP values to identify which meteorological features drive predictions overall. In rainfall prediction, humidity and pressure-related features commonly show high importance, consistent with meteorological understanding. Global SHAP summary plots (to be generated in Python) provide a ranked view of average feature impact.

B. Interpretation of SHAP Visualization

The SHAP visualization illustrates how individual feature values contribute to a specific prediction. Positive SHAP values push the model toward predicting rainfall, while negative values suppress rainfall likelihood. The magnitude reflects contribution strength.

C. Illustrative Instance-Level Explanation

For a correctly predicted *Rain* instance, the model assigns a high rain probability when relative humidity is high, pressure is decreasing, and wind speed is moderate-to-high. SHAP attributions typically show positive contributions from humidity-related variables and negative contributions from stable high-pressure conditions. This aligns with meteorological intuition: moist air and weakening pressure systems increase the likelihood of precipitation.

Even without plotting, reporting the direction of SHAP contributions provides useful transparency for practitioners, indicating whether predictions rely on physically plausible drivers rather than spurious correlations.

Such visual explanations enable domain experts to verify whether the model relies on physically plausible signals, thereby supporting trust and validation prior to deployment.

D. Failure Case / Edge Case (Mandatory)

A representative failure case occurs when the model predicts *No Rain* but the ground truth indicates *Rain*. This often happens under borderline conditions where humidity is elevated but pressure remains relatively stable or wind-related variables provide conflicting signals. In such cases, SHAP typically shows competing contributions: humidity features push the prediction toward rain, while pressure and wind-related features push it away, resulting in a near-threshold probability.

This failure mode is operationally important because missed rainfall events may be costlier than false alarms. The explanation suggests two mitigation directions: (i) incorporate temporal context (e.g., pressure trends over multiple days) and (ii) calibrate decision thresholds to prioritize recall for risk-sensitive settings.

E. Qualitative Validation of Model Reasoning

Explainability provides a qualitative check that model behavior is consistent with meteorological understanding. When humidity is high and pressure decreases, SHAP should typically show positive contributions pushing the prediction toward rain. Conversely, stable high pressure and low humidity should contribute negatively.

F. Confidence and Uncertainty Considerations

Qualitative inspection also helps identify low-confidence cases. Instances where SHAP shows strong competing contributions (some features pushing toward rain and others away) often indicate borderline atmospheric conditions. In operational use, such cases may be flagged for caution or combined with additional information such as temporal trends or NWP outputs.

XVI. Discussion of Practical Implications

The proposed explainable and robust machine learning framework has several practical implications for real-world weather forecasting and decision-support systems. Accurate rainfall prediction is essential for applications such as flood early warning, agricultural planning, water resource management, and transportation safety. By combining strong predictive performance with explainability, the framework supports informed decision-making rather than blind reliance on model outputs.

Explainability plays a critical role in operational deployment. SHAP-based explanations enable meteorologists and domain experts to inspect whether predictions are driven by physically meaningful signals, such as humidity and pressure patterns, or by spurious correlations. This transparency improves trust, facilitates model validation, and helps identify failure modes before deployment in high-stakes environments.

Robustness under dataset shift is another key practical consideration. Weather prediction systems are often trained on historical data collected from specific regions or sensor networks, yet deployed in environments with different climatic conditions or measurement characteristics. The cross-dataset evaluation conducted in this study highlights the importance of testing generalization beyond a single dataset and provides evidence that ensemble-based models, particularly XGBoost, are more resilient to such shifts.

From a system design perspective, the proposed framework can be integrated as a decision-support layer alongside traditional numerical weather prediction models. Rather than replacing physical models, machine learning predictions and explanations can complement existing forecasts, offering additional insight and uncertainty awareness to practitioners.

XVII. Limitations

This study focuses on tabular classifiers and does not explicitly model temporal dependencies in weather sequences. Explainability is primarily assessed via SHAP interpretation rather than formal faithfulness metrics. Cross-dataset evaluation helps, but further testing across multiple regions and seasons would strengthen robustness conclusions.

XVIII. Conclusion and Future Work

This paper proposed an explainable and robust machine learning framework for rainfall prediction using structured meteorological data. By comparing Logistic Regression, Random Forest, and XGBoost within a unified evaluation pipeline, we emphasized predictive performance, cross-dataset robustness, and interpretability through SHAP-based explanations. The framework supports transparent decision-making by highlighting feature contributions at both global and instance levels.

Future work will explore temporal modeling (e.g., recurrent networks and transformer-based architectures) to capture sequential weather dynamics, extend evaluation across additional geographies and station networks, and investigate quantitative measures of explanation quality. Finally, integrating the framework into operational decision-support workflows with stakeholder feedback can further validate trust and utility.

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